

Dexter Fernandes

London • +44 07436007369 • dexterfernandes11@gmail.com • linkedin.com/in/dexterferns •
https://github.com/Dexter-Fernandes

Edge AI/ Computer Vision Engineer

Edge AI / Computer Vision Engineer with experience building real-time video analytics and edge perception systems for industrial safety and inspection environments for clients such as British Petroleum and Austin Industries. Tech stack includes NVIDIA DeepStream video pipelines, TensorRT, OpenCV, Python and C++17, spanning multi-camera ingestion, inference, tracking, monitoring and field debugging. Strong record optimising models and pipelines under latency, compute and memory constraints using FP16/INT8 deployment, profiling and production validation for edge devices. Comfortable owning deployed vision systems end to end, from camera behaviour and model performance through to health checks, structured logging and reliability improvements in 24/7 operation.

PROJECTS

DeepStream RTSP Pipeline

Jun 2026

- Built an end-to-end **NVIDIA DeepStream 9.0/GStreamer RTSP video analytics pipeline** for multi-stream computer vision, using rtspsrc, nvstreammux, nvinfer, nvtracker, and nvdsosd to run YOLO+NvDCF tracking, metadata extraction, anonymisation, CSV logging, and re-streamed output across 3 concurrent camera feeds.
- Deployed and optimised a **YOLO26n TensorRT FP16 detector** on **6GB VRAM Turing hardware**, combining dynamic-batch engine conversion, nvinfer integration, p99 latency profiling, and FP16-vs-FP32 accuracy validation to sustain **29.7 FPS per stream live** and **131.3 FPS per stream unthrottled**.
- Implemented a **C++/CUDA TensorRT decode plugin** using IPluginV2DynamicExt, moving post-processing into the inference engine to support custom-layer deployment.
- Benchmarked tracker configurations across **IOU, NvDCF, and** ByteTrack-inspired **NvSORT**, producing MOTA, IDF1, ID switches, fragmentation, per-frame detection counts, FPS, and VRAM impact to guide production tracker selection.
- Added production hardening through **GPU smoke tests, model-promotion gates, structured JSON logging, and per-camera health metrics** for reactive debugging across liveness, dropped frames, reconnect events, and RSS.

WORK EXPERIENCE

GrowthStage • Remote

May 2026 - Present

Recruiter

- Evaluated candidates' technical depth across perception, SLAM, and deployment using hands-on robotics and computer vision knowledge to separate genuine signal from surface-level claims.

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Developed and maintained NVIDIA DeepStream 6.x video inference pipelines for 24/7 industrial perception deployments, supporting camera ingestion, model inference, and field diagnostics in harsh desert environments.
- Hardened deployed AI services with Dockerised microservices, Kafka messaging, structured logging, health checks, retries, and safe restarts reducing live video pipeline crashes by 67% and improved root-cause analysis.
- Led field validation on BP oil-rig deployments, diagnosing model failures under lighting, motion and environmental stressors and translating findings into practical fixes for live operation.
- Developed a repeatable optimisation workflow (baseline to FP16 to validation) with defined acceptance thresholds, enabling consistent latency, memory, and stability outcomes across deployments.
- Diagnosed and resolved production failures through on-site analysis and structured validation checklists, reducing crash frequency by 67% in live video pipelines.

- Developed an internal web dashboard in Django to surface inference outputs, system health, and performance metrics, enabling faster debugging and root-cause analysis across deployed systems.
- Owned production monitoring and post-deployment validation for deployed vision systems, using observability tooling, failure analysis, and drift tracking to diagnose issues early and improve reliability in live environments.

LivNSense Technologies • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led a team to design, train, and deploy vision pipelines for industrial safety applications, reducing false positives by 33% while meeting latency and power constraints for Austin Industries.
- Built low-latency asynchronous perception pipelines using OpenVINO, and C++17, increasing throughput from 5 FPS to 15 FPS and delivering deterministic real-time inference on Intel hardware.
- Profiled and optimised C++ inference and pre and post-processing hotspots using Intel VTune Profiler for targeted performance fixes in our CNN inference pipeline.
- Ran systematic quantisation and optimisation experiments benchmarking accuracy–latency–memory trade-offs and reducing model size by 50% while meeting real-time constraints.
- Collaborated with hardware, software, and QA teams to integrate CV models into deployed systems, and presented performance findings to leadership and clients to support rollout, optimisation, and reliability decisions.
- Mentored junior engineers and collaborated with hardware, software, and QA teams to integrate perception modules into deployed systems and improve delivery consistency.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Dissertation: designed and implemented a Radiological SLAM system on a Clearpath Husky platform, integrating LiDAR and radiological data for localisation, mapping, and inspection workflows with Python.
- Validated sensor placement for coverage and calibrated LiDAR extrinsics to the robot base, VLP-16 and Intel RealSense D435i.
- Performed a JetPack platform upgrade on Jetson Xavier AGX to access newer CUDA and TensorRT, porting a ROS2/PCL SLAM stack across the upgrade and resolving the dependency and driver breakage it introduced.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Built a strong foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, sensor systems, and robotics software.

SKILLS

- **Edge AI & Video Pipelines:** NVIDIA DeepStream 6.x/9.0, GStreamer, RTSP video pipelines, nvstreammux, nvinfer, nvtracker, nvdssd, CUDA, JetPack, camera behaviour debugging, optical performance eval, ISP tuning
- **Edge Inference and Optimisation:** FP16/INT8 deployment, OpenVINO, TensorRT, ONNX, custom TensorRT plugins, dynamic batching, latency/throughput profiling
- **Computer Vision & Image Processing:** Object detection, Object tracking, RT-DETR, YOLOv8, YOLO26, SSD MobileNet V2, EfficientDet, NvDCF, ByteTrack
- **Cloud, Edge & MLOps:** Apache Kafka, MLflow, Docker, CI/CD, Production monitoring, Structured logging, Health checks, model-promotion gates, failure-mode analysis
- **Programming & Tooling:** Python, C++17, Bash, Django, CMake, Git, Intel VTune, OpenCV, CUDA C++
- **Edge AI Hardware:** NVIDIA Jetson AGX Xavier, Intel Core i7 (VTC7252-7C4IP), Intel Movidius Myriad VPU